3.1.2 Monotone Sequences

Definition: A sequence $\{a_n\}_{n=1}^{\infty}$

- is called increasing if
- is called decreasing if
- A sequence that is increasing or decreasing is called
- A sequence that is strictly increasing or strictly decreasing is called

Example 1: Prove that the sequence $a_n = \dfrac{n}{n+1}$ is strictly increasing.

Definition: If discarding finitely many terms from the beginning of a sequence produces a sequence with a certain property, then the original sequence is said to have that property eventually.

Example 2:

Show that the sequence $a_n = \left\{ \frac{10^n}{n!} \right\}_{n=1}^{\infty}$ is eventually strictly decreasing.

Definition: A sequence $\{a_n\}$ is

A sequence $\{a_n\}$ is

A sequence $\{a_n\}$ is

Example 3: Show that the sequence $a_n = 6 + \frac{1}{n}$ is a bounded sequence

Theorem (Monotonic Sequence Theorem):

Every bounded above, increasing (or eventually increasing) sequence is convergent.
Every bounded below, decreasing (or eventually decreasing) sequence is convergent.

More often stated as:
Every bounded, monotone (or eventually monotone) sequence is convergent.

The proof of this theorem requires some additional content knowledge about the Completeness Axiom. We will not discuss the proof in class, however details can be found on pages 611 & 612 of the textbook.

Note: This theorem is very valuable in establishing whether a sequence is convergent. It says nothing about what a sequence converges to.

Example 4 (you try): Consider the sequence $a_n = \frac{1}{n}$

(a) Prove that this sequence is decreasing.
(b) Show that this sequence is bounded below.
(c) What can you conclude from parts (a) and (b) by the Monotonic Sequence Theorem?

Example 5: Show that the sequence $a_n = \left\{ \frac{10^n}{n!} \right\}_{n=1}^{\infty}$ converges and find its limit.